

Tianyu (Anna) Yang

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Overview

PhD candidate in Computer Science with 15 publications at top ML/NLP venues (ACL, EMNLP), including **8 first author**. Applied this research as a research intern at Zillow, Samsung Research, and Bosch AI Center, building continual-alignment, RL-based long-term memory consolidation, and agentic multimodal RAG systems. **Research focus: post-training LLMs/MLLMs for multimodal agents that perceive, reason, and act.**

Education

Ph.D., Computer Science University of Notre Dame Advisor: Prof. Xiangliang Zhang; GPA: 3.90/4.0	Sep 2023 – May 2027 (expected) South Bend, IN
M.S. (Research), Electrical Engineering Columbia University Advisor: Prof. Shih-Fu Chang; GPA: 3.90/4.0	Sep 2021 – May 2023 New York, NY
B.S., Automation Beijing Jiaotong University GPA: 3.70/4.0	Sep 2017 – May 2021 Beijing, China

Work Experience

Zillow Applied Scientist Intern - Investigating alignment drift in self-evolving LLMs, where continual capability acquisition can introduce new safety risks and degrade previously learned alignment. - Developing Alignment Sleep, a continual alignment framework for detecting and repairing safety drift while preserving helpfulness and newly acquired capabilities.	Jun 2026 – Present Seattle, WA
Samsung Research Research Scientist Intern - Developed TrustMem, an RL framework for trustworthy long-term memory consolidation in LLM agents. - Designed a Memory Transition Verifier to evaluate memory updates for coverage, preservation, and faithfulness. - Achieved state-of-the-art performance across major long-term memory benchmarks, improving MemoryAgent-Bench by 6.5 points and HaluMem by 12.14 F1 points while reducing memory errors by up to 79.1%. <i>Resulting publication: TrustMem: Learning Trustworthy Memory Consolidation for LLM Agents with Long-Term Memory (Under Submission)</i>	Jan 2026 – May 2026 Mountain View, CA
Bosch AI Center Research Scientist Intern - Developed MM-R2, an agentic multimodal RAG framework that reasons about what to retrieve and where to search before retrieval. - Designed intent-referent binding and KnowledgeMap routing for grounded and interpretable evidence retrieval. - Built MM-R2-Traj with 900K retrieval trajectories and trained the agent with SFT and GRPO, outperforming the strongest prior method by 7.8 points on InfoSeek. <i>Resulting publication: Reason Before You Retrieve: Agentic Planning for Multi-modal RAG (Under Submission)</i>	May 2025 – Aug 2025 Pittsburgh, PA

Academic Research & Projects

LLM Agents for Multimodal Scientific Data Collection and Synthesis University of Notre Dame Advisor: Prof. Xiangliang Zhang	Feb 2025 – Aug 2025 South Bend, IN
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- Developed Quest2DataAgent, an AI-for-Science multi-agent system that automates end-to-end scientific data discovery and collection from natural-language research questions across ecology and polymer science.
- Built LLM agents for planning, hybrid Neo4j-FAISS retrieval, evaluation, and visualization over heterogeneous multimodal data, reducing data collection from hours or days to minutes; selected for an NSF-funded initiative supporting partner universities.

Multimodal Machine Unlearning with CLIP for Cross-Modal Alignment May 2024 – Oct 2024
University of Notre Dame South Bend, IN

Advisor: Prof. Xiangliang Zhang and Prof. Yapeng Tian

- Proposed CLIPERase, one of the first multimodal machine unlearning methods for removing sensitive visual-textual associations from CLIP while preserving cross-modal alignment.
- Designed modular forgetting, retention, and consistency objectives without retraining from scratch, demonstrating broad effectiveness across five downstream tasks including retrieval, image generation, and MLLMs.

Semantic Audio-Visual Representation Learning for Question Answering Sep 2023 – Feb 2024
University of Notre Dame South Bend, IN

Advisor: Prof. Xiangliang Zhang and Prof. Yapeng Tian

- Developed SaSR-Net, a source-aware audio-visual representation learning framework that aligns sound sources, visual objects, and question semantics for fine-grained multimodal reasoning.
- Introduced source-wise learnable tokens and spatial-temporal attention, achieving state-of-the-art performance on Music-AVQA and AVQA-Yang with moderate computational cost.

Google Summer of Code: G-API Algorithms for OpenCV May 2023 – Aug 2023
Google Remote

- Developed Python tutorials and new machine learning algorithms for OpenCV’s G-API module, extending official documentation and improving image-processing performance.

Publications

Preprints and Under Submission

- [Sub.] [TrustMem: Learning Trustworthy Memory Consolidation for LLM Agents with Long-Term Memory](#)
Tianyu Yang, Sudipta Paul, Vijay Srinivasan, Vivek Kulkarni, Srinivas Chappidi
- [Sub.] [Reason Before You Retrieve: Agentic Planning for Multi-modal RAG](#)
Tianyu Yang, Minghao Chen, Zhenzhen Li, Simon Shir, Xiangliang Zhang
- [Sub.] [Agentic Multimodal RAG: Roles, Decisions, and Evaluation](#)
Tianyu Yang*, Wenliang Guo*, Xiujin Liu, Shijian Deng, Yilun Zhao, Yapeng Tian, Xiangliang Zhang
- [Sub.] [DesignAgent: Interactive Object-Level Indoor 3D Scene Editing via Multimodal Agentic Reasoning](#)
Tianyu Yang*, Xiujin Liu*, Chen Zhao, Yilun Zhao, Xiangliang Zhang

Selected Peer-Reviewed Publications

- [ACL’26] [Deconstructing Multimodal Mathematical Reasoning: Towards a Unified Perception-Alignment-Reasoning Paradigm](#)
Tianyu Yang, Sihong Wu, Yilun Zhao, Lisen Dai, Chen Zhao, Minhao Cheng, Arman Cohan, Xiangliang Zhang
- [EMNLP’25] [Quest2DataAgent: Automating End-to-End Scientific Data Collection](#)
Tianyu Yang, Jason Rohr, Brett Scheffers, Nitesh Chawla, Xiangliang Zhang
- [ACL’25] [CLIPERase: Efficient Unlearning of Visual-Textual Associations in CLIP](#)

Tianyu Yang, Lisen Dai, Xiangqi Wang, Minghao Chen, Yapeng Tian, Xiangliang Zhang

- [EMNLP'24] **SaSR-Net: Source-Aware Semantic Representation Network for Enhancing Audio-Visual Question Answering**
Tianyu Yang, Yiyang Nan, Lisen Dai, Zhenwen Liang, Yapeng Tian, Xiangliang Zhang
- [COLM'26] **Explainable AI-Generated Image Detection RewardBench**
Michael Yang, Shijian Deng, William T. Doan, Kai Wang, **Tianyu Yang**, Harsh Singh, Yapeng Tian
- [WSDM'25] **WildlifeLookup: A Chatbot Facilitating Wildlife Management with Accessible Data and Insights**
Xiangqi Wang, **Tianyu Yang**, Jason Rohr, Brett Scheffers, Nitesh Chawla, Xiangliang Zhang
- [EMNLP'25] **Self-Improvement in Multimodal Large Language Models: A Survey**
Shijian Deng, Kai Wang, **Tianyu Yang**, Harsh Singh, Yapeng Tian
- [ACL'25] **Physics: Benchmarking Foundation Models for PhD-Qualifying Exam Physics Problem Solving**
Kaiyue Feng, Yilun Zhao, **Tianyu Yang**, Yixin Liu, Chen Zhao, John Sous, Arman Cohan
- [ACL'25] **Beyond Single-Value Metrics: Evaluating and Enhancing LLM Unlearning with Cognitive Diagnosis**
Yicheng Lang, Kehan Guo, Yue Huang, Yujun Zhou, Haomin Zhuang, **Tianyu Yang**, Yao Su, Xiangliang Zhang
- [ACL'24] **SceMQA: A Scientific College Entrance Level Multimodal Question Answering Benchmark**
Zhenwen Liang, Kehan Guo, Gang Liu, Taicheng Guo, Yujun Zhou, **Tianyu Yang**, Jiajun Jiao, Renjie Pi, Jipeng Zhang, Xiangliang Zhang
- [EMNLP'23] **UniMath: A Foundational and Multimodal Mathematical Reasoner**
Zhenwen Liang, **Tianyu Yang**, Jipeng Zhang, Xiangliang Zhang

* *Co-first authors.*

Program Committee Member

- **2026** – Area Chair, ARR Rolling Review
- **2026** – Reviewer, ACL, KDD, CVPR, ECCV, COLM, NeurIPS, ICML, ICLR
- **2025** – Program Chair / Organizer, AAAI Workshop: AI for Science
- **2025** – Program Chair / Organizer, ICCV Workshop: Knowledge-Intensive Multimodal Reasoning
- **2025** – Organizer, CVPR Tutorial: Multimodal Mathematical Reasoning
- **2025** – Reviewer, NAACL, ACL, NeurIPS, EMNLP, CVPR, ICCV
- **2024** – Reviewer, EMNLP, ACL, ICDM

Skills

Research Methods: LLM Post-training (SFT, RLHF, DPO, PPO, GRPO), Reward Modeling & Preference Optimization, Agentic RL, Retrieval-Augmented Generation (RAG) & Agentic RAG, Tool-Use & Multi-Agent Systems, Long-Context Memory Agents, Multimodal Reasoning, Model Evaluation & Benchmarking

ML/LLM Frameworks: PyTorch, Hugging Face Transformers/Datasets, TRL, verl, PEFT/LoRA/QLoRA, DeepSpeed, Accelerate, Ray, vLLM, FAISS, Neo4j

Model Efficiency: Multi-GPU Distributed Training, Model Quantization (INT8/INT4), Knowledge Distillation
Engineering & Tooling: Python, C/C++, SQL, Bash, Linux, Git, Docker, uv, Slurm, HDFS, Streamlit